Level-1-Data Analytics and Predictive Modeling Project Documentation

Project Title:

Supply Chain Optimization

Team Members:

- Madhavendra Singh Naruka

Project Overview:

The goal of this project is to optimize the supply chain operations of a company to reduce costs, improve efficiency, and enhance customer satisfaction. This will involve analysing the current supply chain processes, identifying bottlenecks and inefficiencies, and developing strategies and models to optimize the overall supply chain performance.

1. Introduction

- **Purpose**: The goal of this project is to optimize the supply chain operations of a company to reduce costs, improve efficiency, and enhance customer satisfaction.
- **Scope**: Project will involve analysing the current supply chain processes, identifying bottlenecks and inefficiencies, and developing strategies and models to optimize the overall supply chain performance.

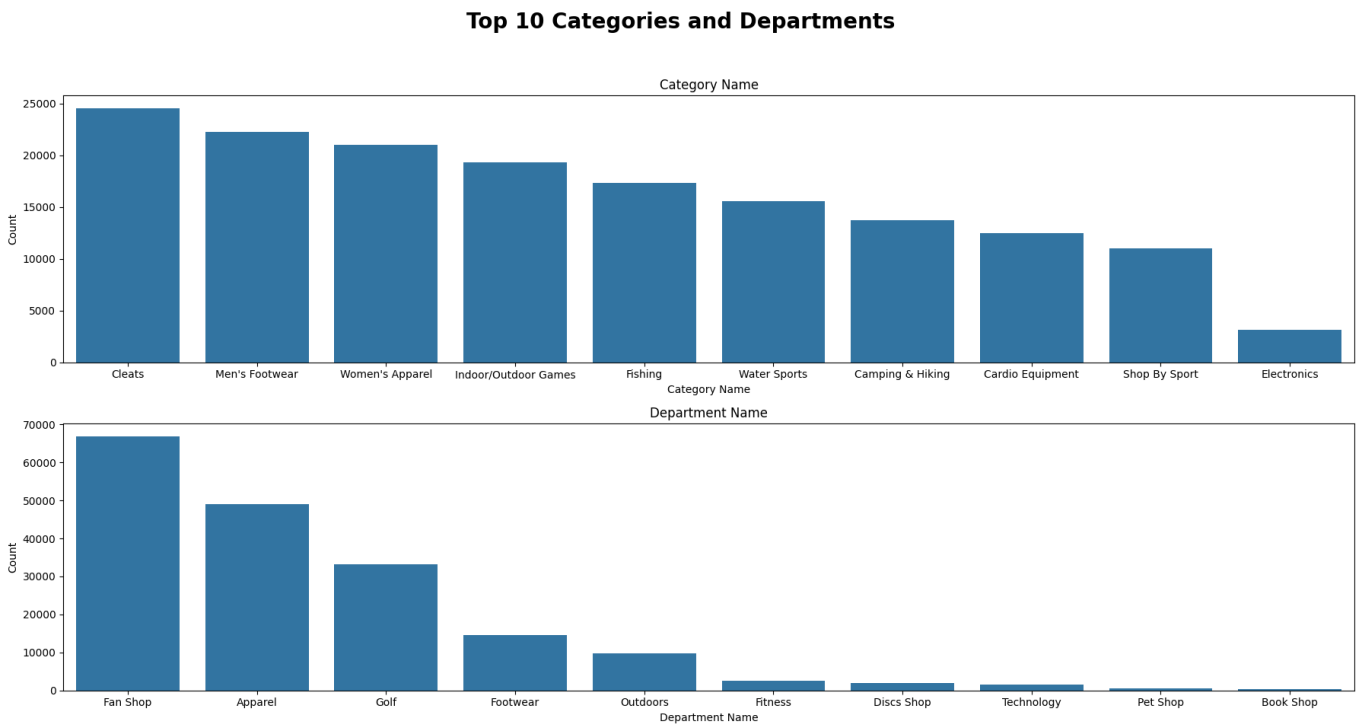
2. Data Collection

- **Data Sources**: I downloaded data form Kaggle https://www.kaggle.com/datasets/shashwatwork/dataco-smart-supply-chain-for-big-data-analysis .
- **Data Information**: The data is pre-processed and structured and has historical record of over 1.4 lakh orders.

3. Data Preprocessing

- As I worked through the project I realized that dataset did not have a few crucial columns for complete Analysis so I created those columns like Post_Order_Action, Transportation_Cost and Supplier IDs, etc.
- I removed several unnecessary columns to simplify dataset including personal information like email, password ,image and description etc.

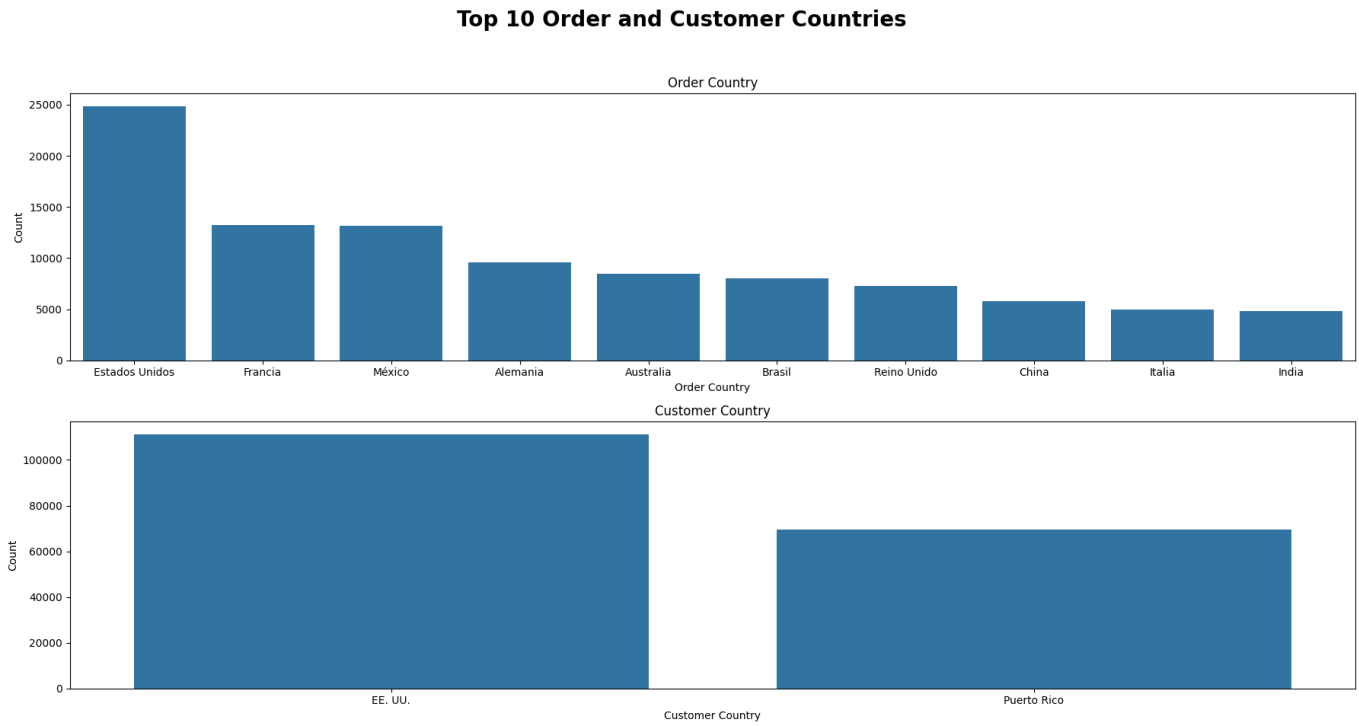
4. Exploratory Data Analysis (EDA)



Significant Findings:

- **Categories**: The distribution of counts among the top categories is relatively even, indicating less variance.
- **Departments**: There is significant variance in the counts among the top departments, with a few departments (Fan Shop, Apparel, and Golf) comprising a major part of the activity or sales.

These findings suggest that while interest or sales are more evenly spread across the top categories, certain departments dominate the activity or sales, indicating areas of higher focus or demand. This insight can be useful for targeted marketing, inventory management, and strategic planning in your internship project.

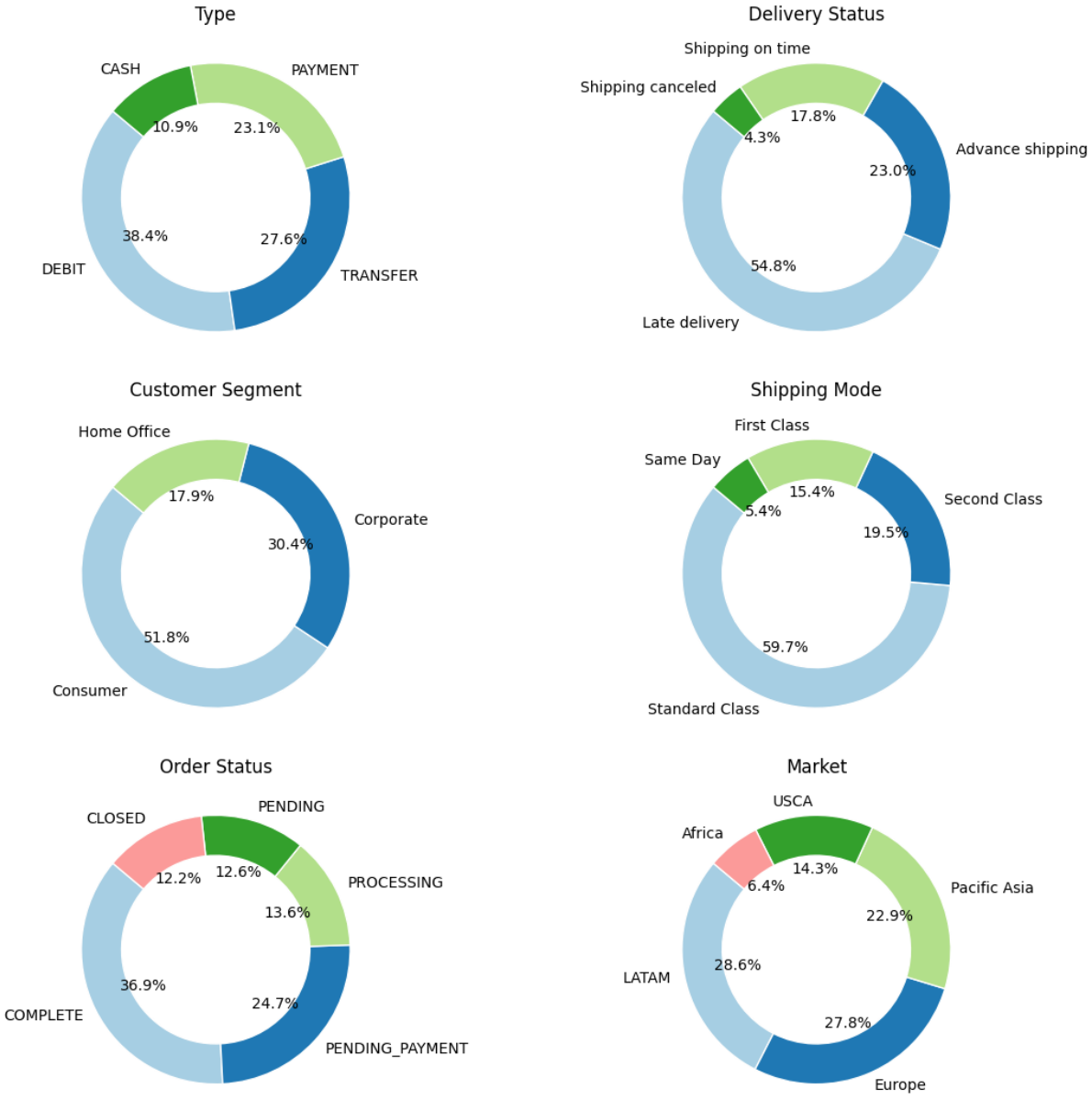


Significant Findings :

- **Order Countries**: The counts are relatively more spread out, with Estados Unidos leading significantly but a more gradual decline among the top 10 countries.
- **Customer Countries**: There is a significant variance, with EE. UU. having a much higher count compared to Puerto Rico.

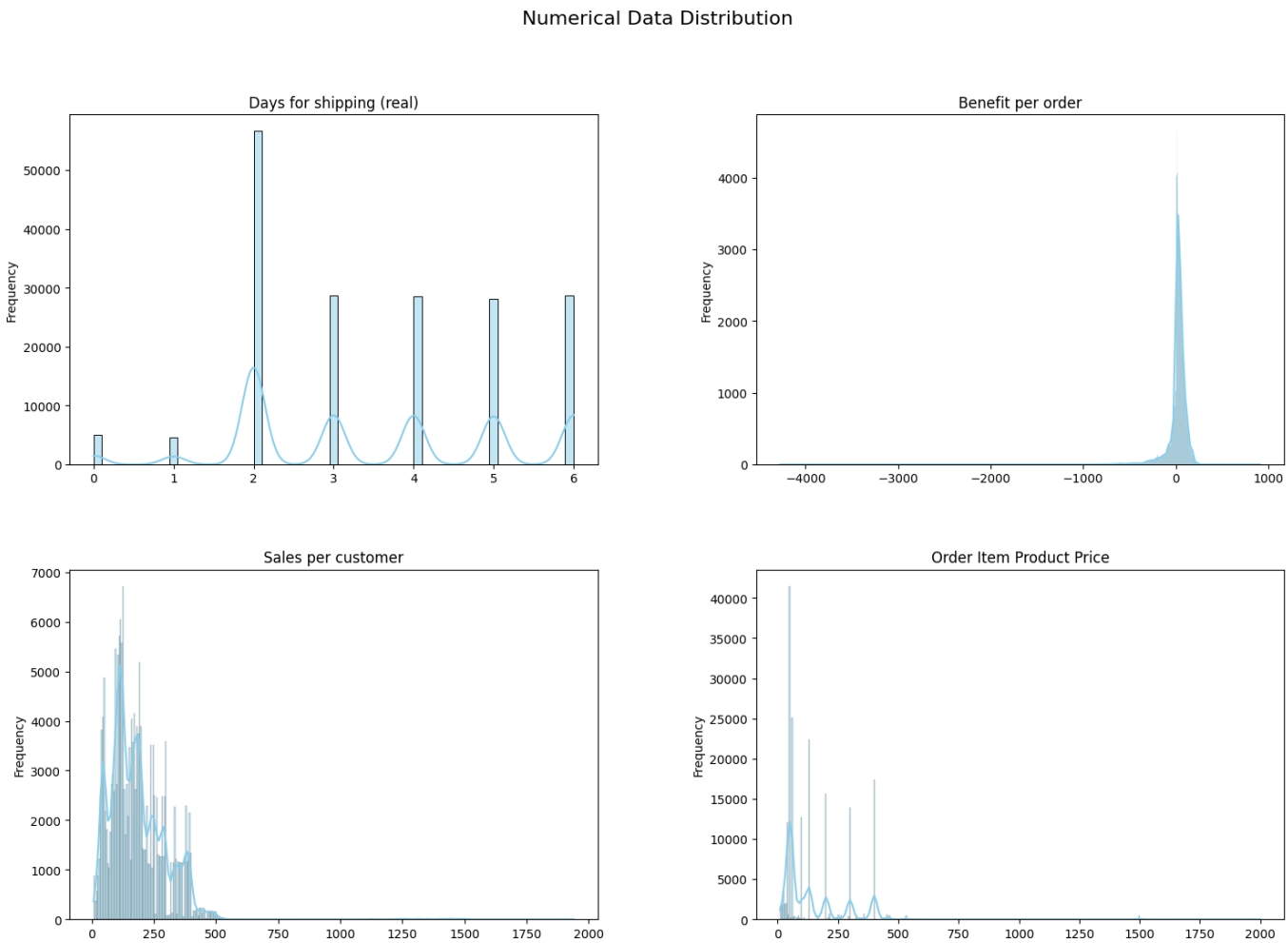
These insights can help in understanding the major markets and customer bases, which is valuable for strategic planning in targeting specific regions, optimizing logistics, and tailoring marketing efforts for your internship project.

Categorical Data Distribution



Significant Findings:

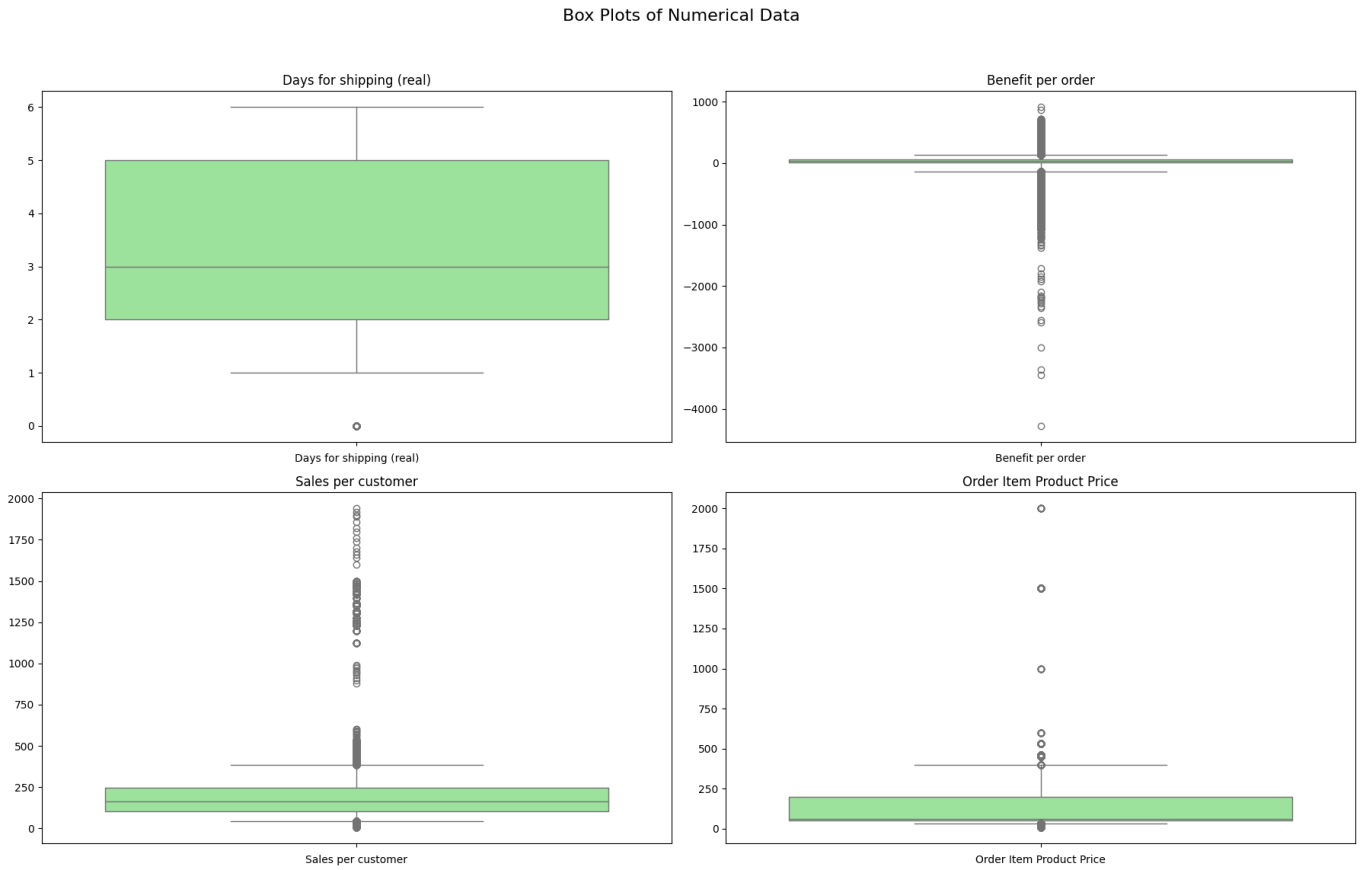
- **Type**: DEBIT and TRANSFER are the most common payment types, making up over half of the total.
- **Delivery Status**: Late delivery is the most frequent, indicating potential issues in the delivery process.
- **Customer Segment**: Consumers represent the largest customer segment, indicating a strong consumer market.
- **Shipping Mode**: Standard Class is the predominant shipping mode, suggesting most customers opt for standard shipping over faster options.
- **Order Status**: COMPLETE orders are the most common, indicating efficient order processing.
- **Market**: LATAM and Europe are the largest markets, showing strong regional demand.



Significant Findings:

- **Days for Shipping (Real):**The most common shipping time is 2 days. There are distinct peaks at each day from 0 to 6, indicating consistent shipping times with little variance within each day.
- **Benefit per Order:** The distribution is highly concentrated around zero, with a significant number of orders showing little to no profit. There are some negative values, indicating that certain orders resulted in losses.
- **Sales per Customer:** The distribution shows a right skew, with most sales per customer being low, primarily below 500 units. There is a long tail, indicating that a small number of customers make significantly larger purchases.
- **Order Item Product Price:** The distribution is right-skewed, with most products priced on the lower end. A few high-priced items exist, contributing to the long tail in the distribution.

Better Pricing and eliminating products which result in frequent losses is recomended



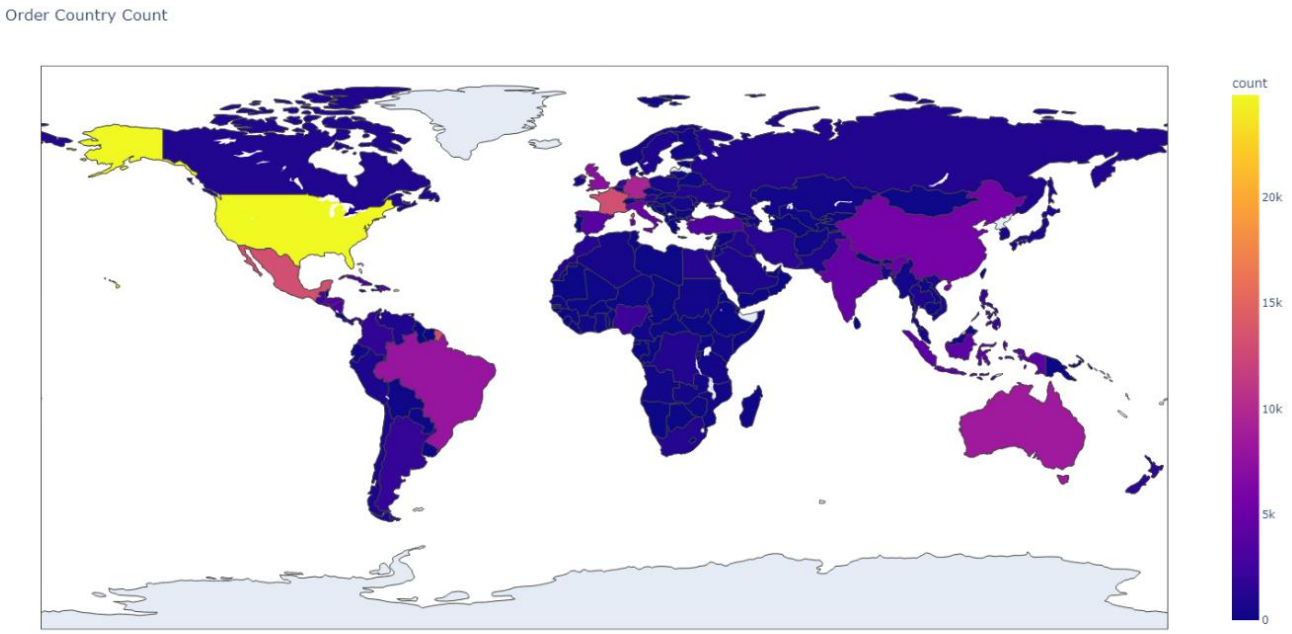
Significant Findings from the Box Plots

1. **Days for Shipping (Real):**
   - The median shipping time is around 3 days.
   - The interquartile range (IQR) is between 2 and 5 days, indicating that the middle 50% of orders are shipped within this range.
   - There is a lower outlier at 0 days, suggesting some immediate or very fast shipping occurrences.
2. **Benefit per Order:**
   - The median benefit per order is slightly below zero, suggesting that half of the orders result in little to no profit.
   - The IQR is narrow, indicating that most benefits are close to zero.
   - Numerous outliers below the lower whisker show significant losses on some orders, while there are a few positive outliers as well.
3. **Sales per Customer:**
   - The median sales per customer are low, around 50 units.
   - The IQR is between approximately 25 and 125 units, showing that most customers purchase within this range.
   - There are several outliers above 250 units, indicating that some customers make much larger purchases.
4. **Order Item Product Price:**
   - The median product price is relatively low, around 100 units.

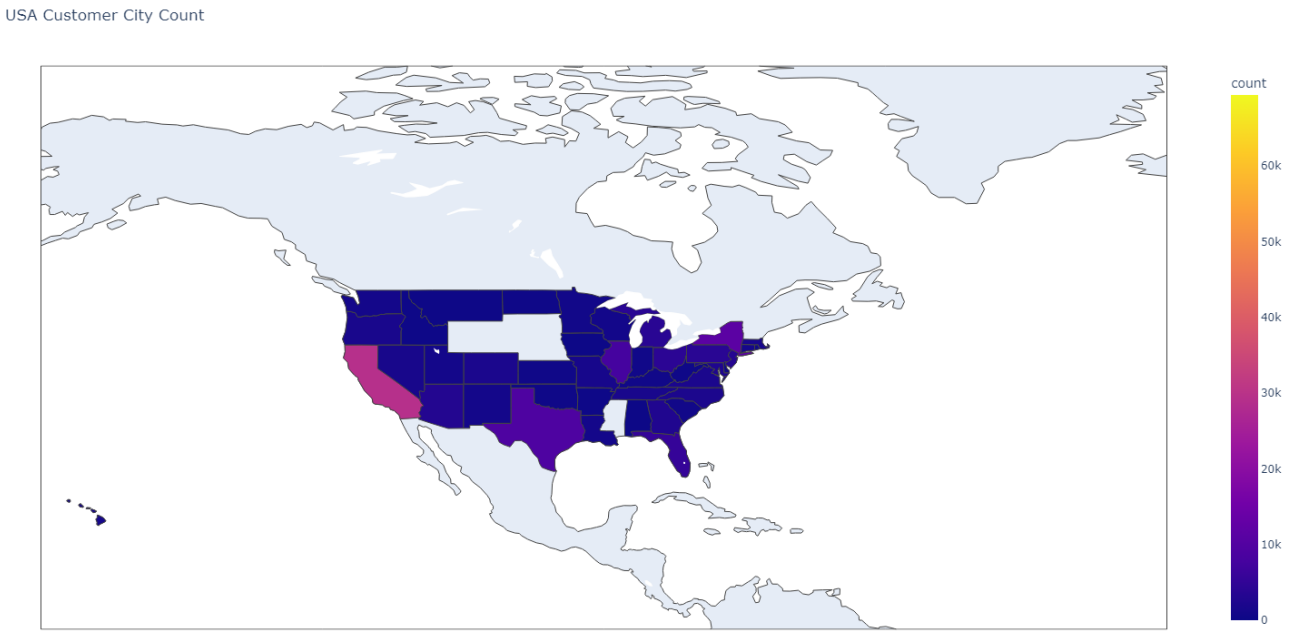
- The IQR ranges from approximately 50 to 200 units, indicating most products are priced within this range.
- A few high-priced items are present, as shown by the outliers above 500 units.

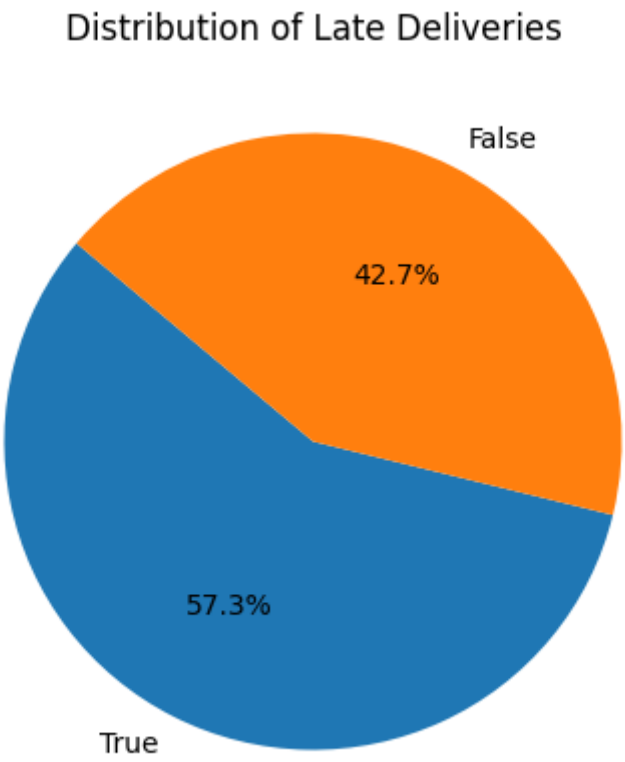
Recommendations: To improve profitability, consider reviewing and optimizing pricing strategies, particularly for products contributing to losses. Additionally, identifying and catering to high-value customers could help increase sales per customer.

Distribution of Countries from where products were purchased/shipped



Distribution of Cities from where products were ordered

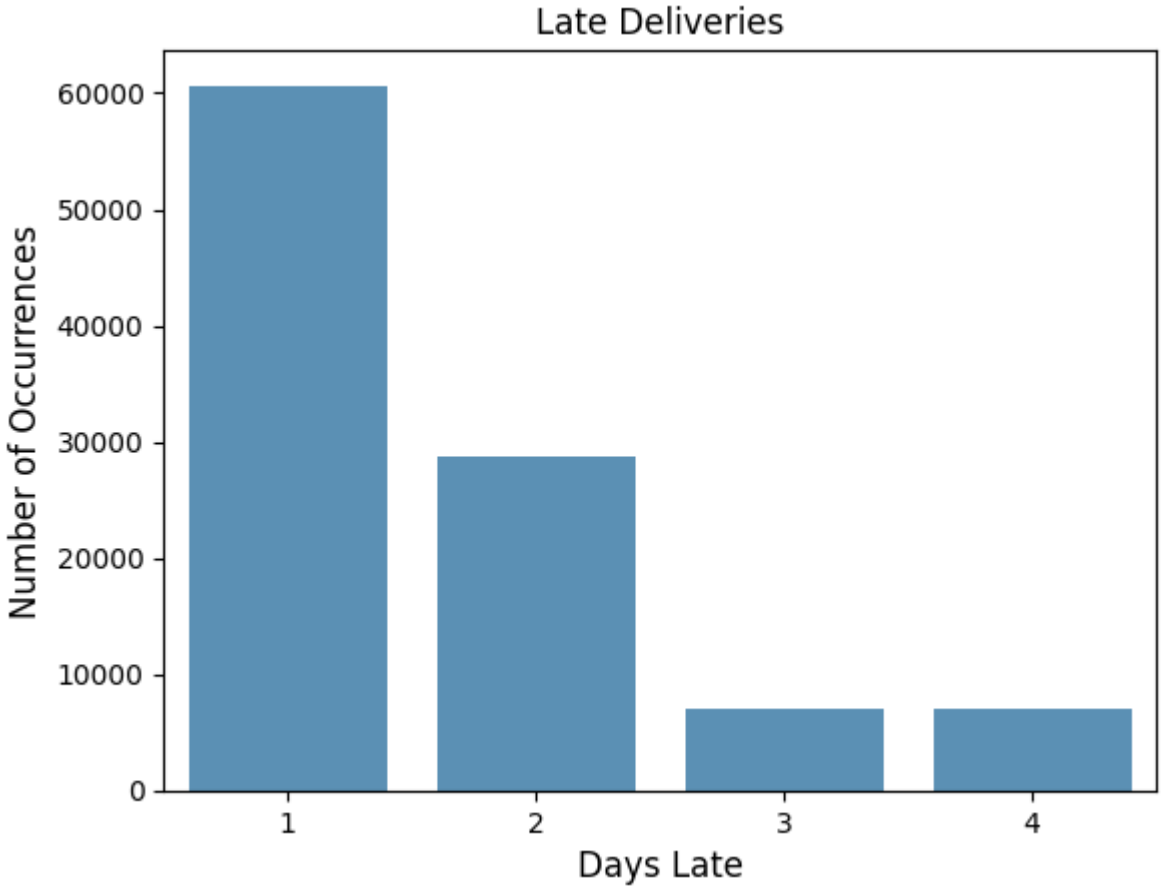




Significant Findings

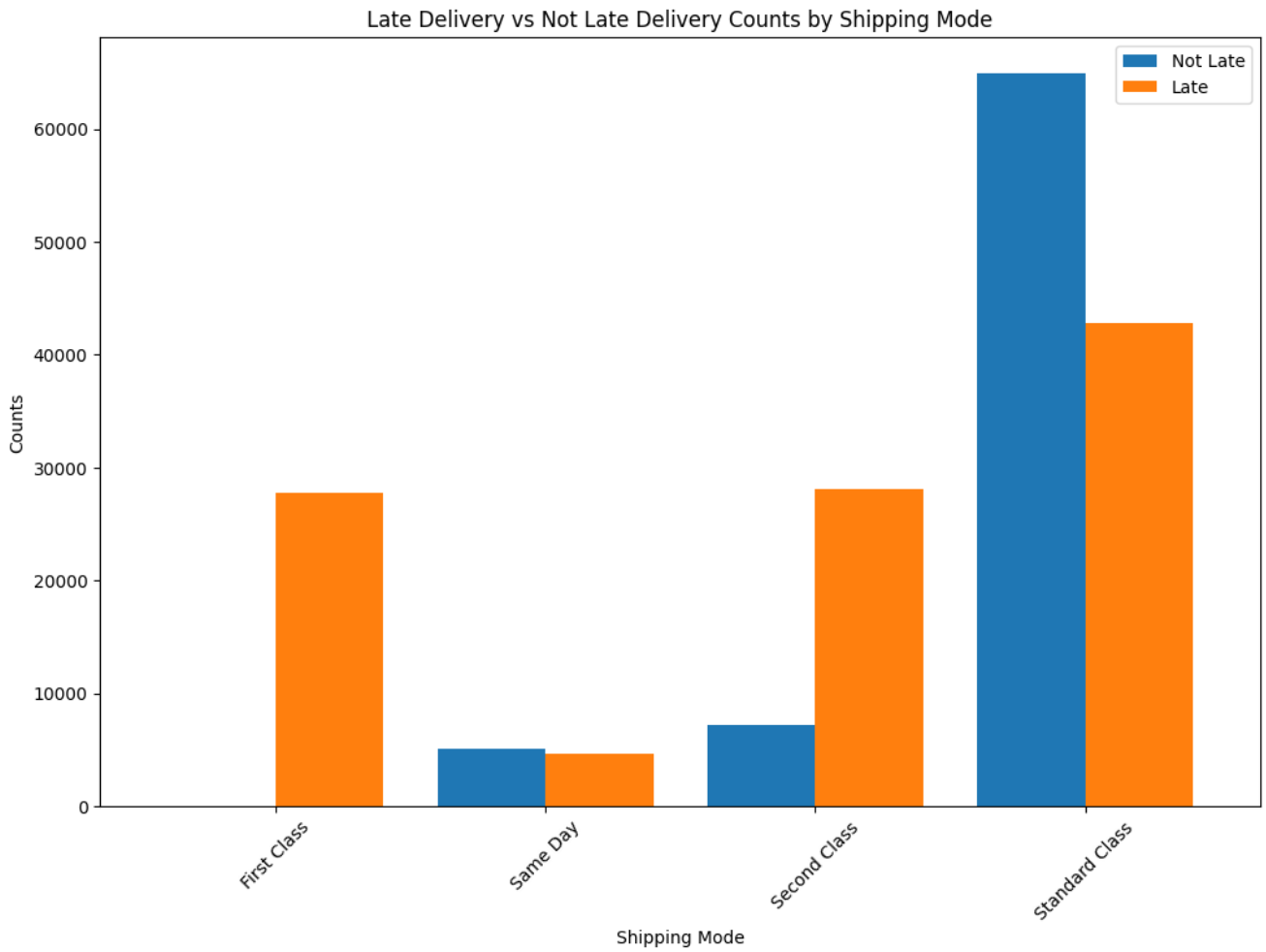
- Large amount of deliveries are Late, optimizing Supply chain is recommended to ensure on-time delivery.
- I also calculated the fill-rate which tell us the rate at which orders were delivered on time. Delivery routes need to be optimized to ensure customer satisfaction.

	Total Orders	On Time Orders	Fill Rate
0	180519	73788	40.88%



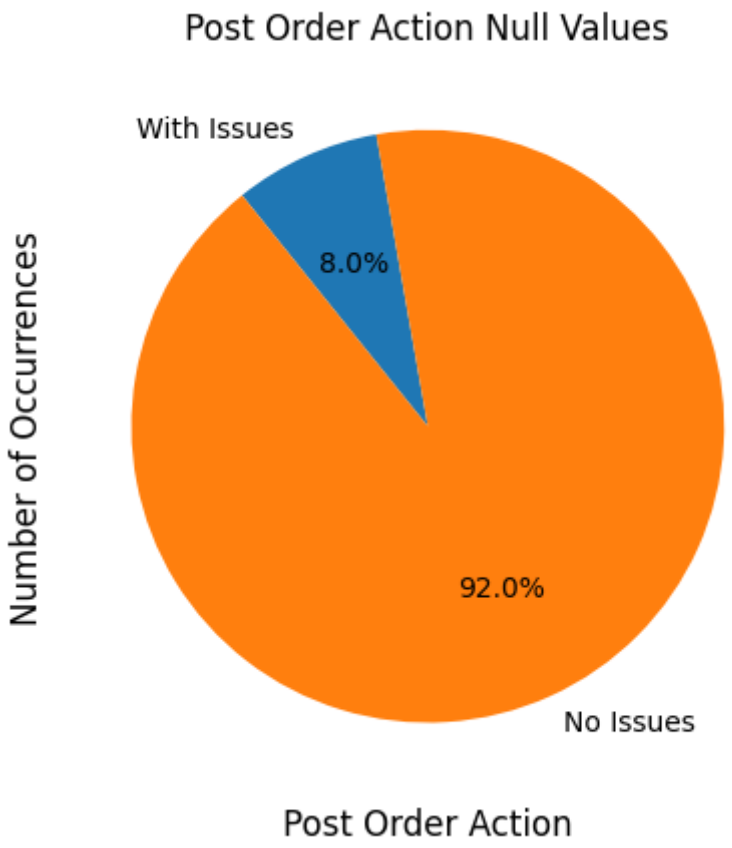
Significant Findings :

- Most late deliveries are one or two days late.



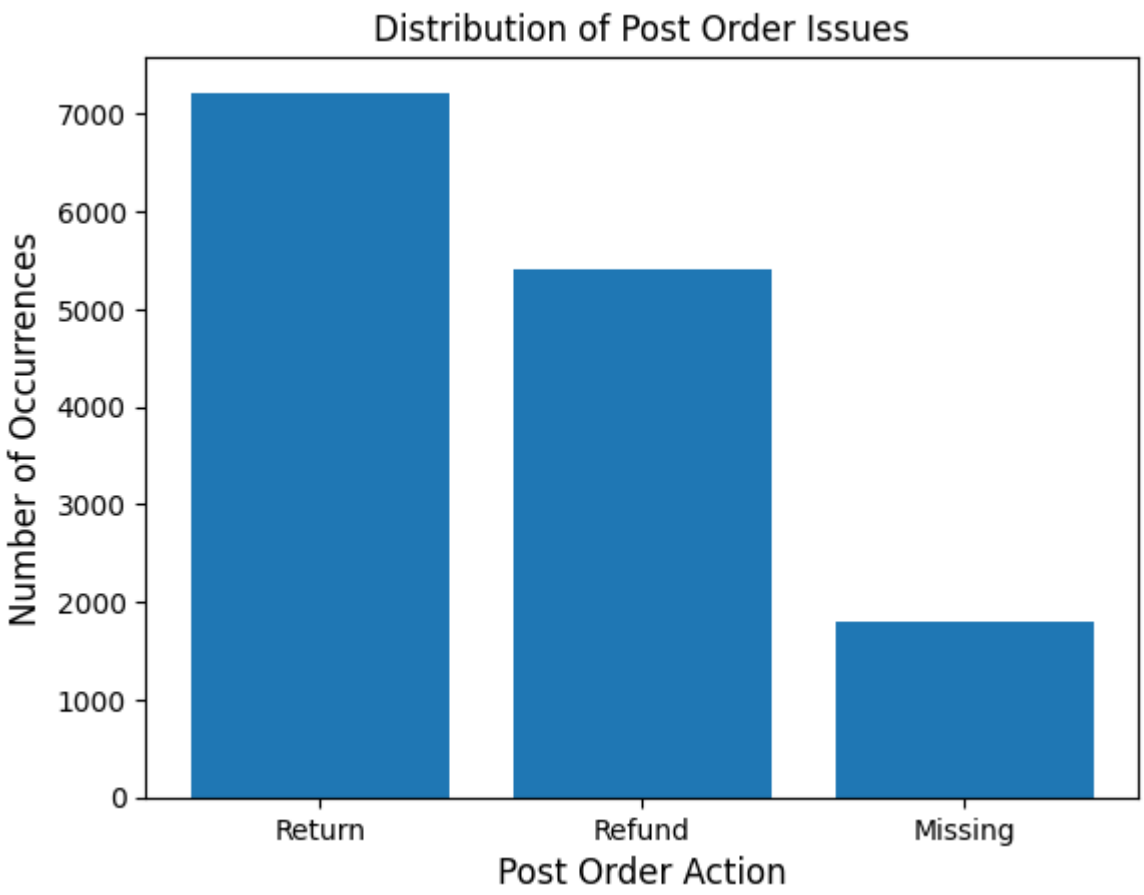
Significant Findings :

- Standard deliveries experience fewer delays compared to premium modes.
- The substantial delays in premium modes may contribute to their low usage.
- Addressing on-time delivery issues across all modes can enhance the company's reputation.
- In the mean time, the Estimated Delivery Time should be adjusted to account for potential delays. This proactive approach fosters customer satisfaction and builds trust.



Significant Findings :

- There are less issues but the can add up. So further analysis to understand the issues should be done

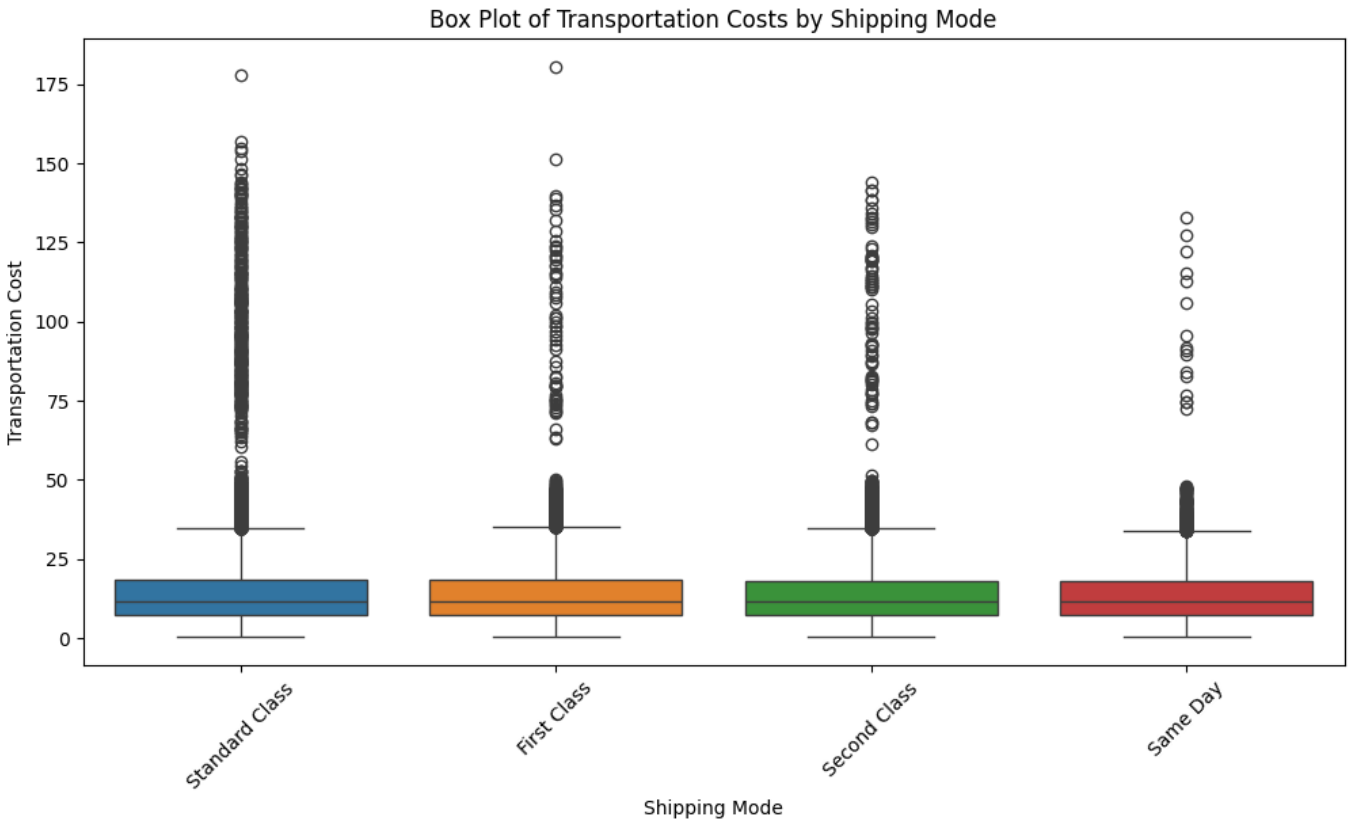


Significant Findings :

- Return and Refund are major post order action in terms of Order issues
- The number of orders going missing is alarmingly high as they will tarnish the brand reputation most.

Significant Findings :

- Transportation costs seems evenly spread out across different Prices over all Shipping Modes.

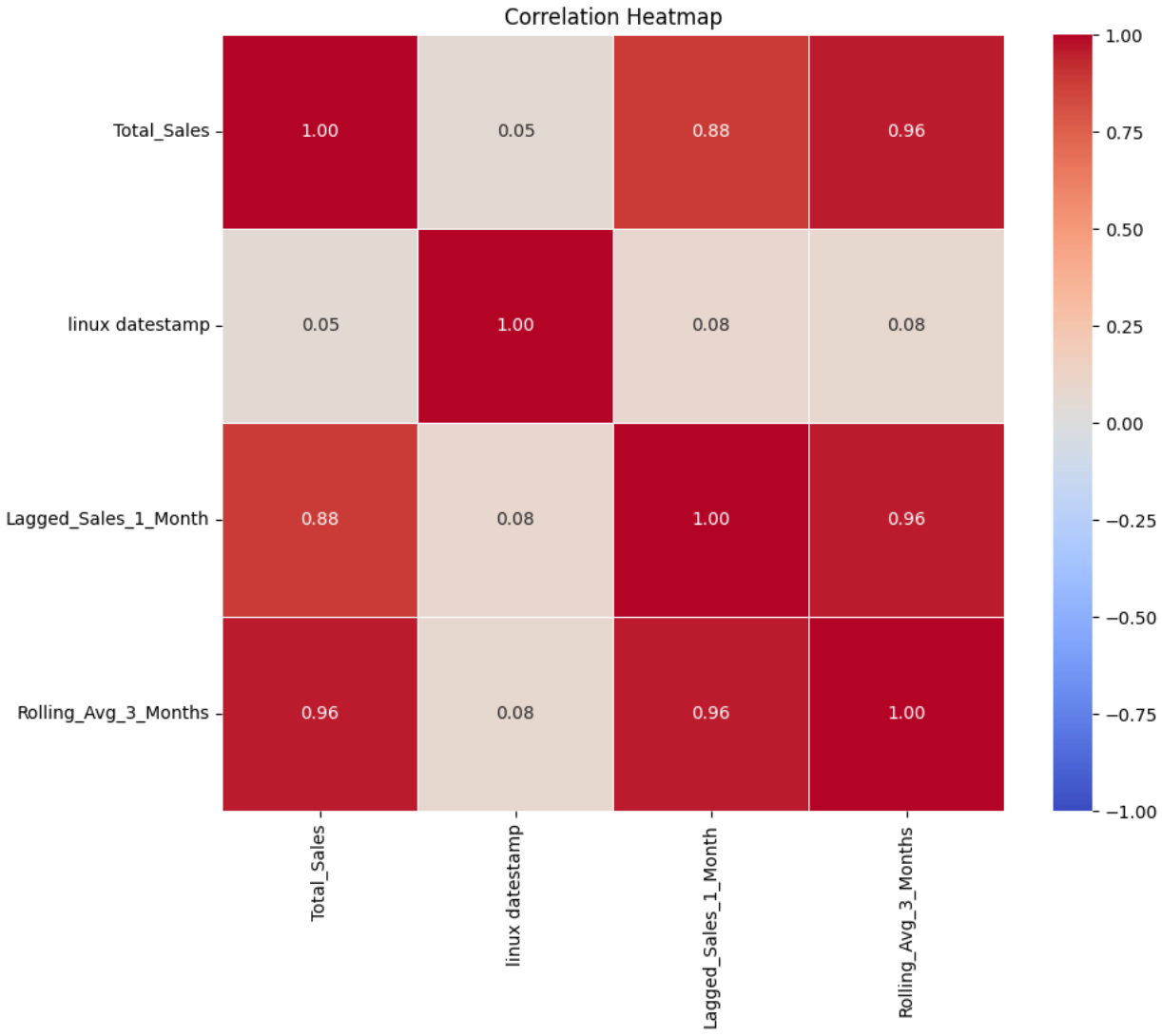


Significant Findings:

- The box indicates that 50% of orders costs lie around 10 to 20 points for all modes while there is variable variance of outliers for each mode.

5.Feature Engineering

- **Feature Selection**: I selected 'Total_Sales' and 'Rolling_Avg_3_Months' as features for my demand forecasting models as they showed highest correlation with Total Sales.



- **Feature Creation**: I have created several features like Lead Time, Fill Rate, order Accuracy, Lagged Sales 1 month, Rolling Average 3 months, EOQ and JIT, Z-score, Safety Stock.
    - **Lead Time**: The time it takes from order placement to product delivery.
    - **Fill Rate**: Measures how well customer orders are fulfilled.
    - **Order Accuracy**: Ensures correct and error-free order processing.
    - **Lagged Sales (1 month)**: Considers sales data from the previous month.
    - **Rolling Average (3 months)**: Smooths out fluctuations by averaging sales over three months.
    - **EOQ (Economic Order Quantity)**: Determines optimal order quantity to minimize costs.
    - **JIT (Just-In-Time)**: Aims to reduce inventory by ordering only when needed.
        - Reorder Point (ROP): The inventory level at which a new order should be placed to prevent stockouts. It ensures that stock is replenished before it runs out.
        - Lead Time: The time required from placing an order to receiving it. Effective management of lead time ensures timely replenishment and aligns inventory levels with demand.
        - Daily Demand: The average demand per day, calculated from the total demand rate. It helps in setting appropriate reorder points and determining how much inventory is needed each day.
    - **Z-score**: The **Z-score** is calculated by subtracting the mean from an individual data point and dividing the result by the standard deviation. It's used to assess how far a data point deviates from the average and helps evaluate supplier performance or identify outliers in a dataset.
    - **Safety Stock**: Extra inventory to handle unexpected demand fluctuations using Z-score.

Level-2-Data Analytics and Predictive Modeling Project Documentation

Project Title:

Supply Chain Optimization

Team Members:

- Madhavendra Singh Naruka

6. Demand Forecasting Model Development

- **Model Selection**: I selected various models like
  - Linear Regression: Simple and interpretable; assumes a linear relationship between features and demand.
  - Decision Tree: Captures non-linear patterns; easy to visualize and understand.
  - Random Forest: Ensemble of decision trees; robust and handles complex interactions.
  - Support Vector Machine (SVM): Effective for small datasets; finds optimal hyperplanes.
  - Gradient Boosting: Combines weak learners into a strong model; minimizes prediction errors.
- **Model Training**: I used a for loop to train various models and store their evaluation metrics.
- **Model Evaluation**:

	Model	MSE	RMSE	R2	MAE
0	Linear Regression	3.810867e+08	19521.441811	0.887948	4231.383571
1	Decision Tree	6.958776e+08	26379.491582	0.795389	7781.764550
2	Random Forest	6.630920e+08	25750.572123	0.805029	7470.391887
3	Support Vector Machine	4.087408e+09	63932.842690	-0.201833	26887.806859
4	Gradient Boosting	6.647984e+08	25783.684202	0.804527	7249.947348

The R2 score for Linear Regression is highest which is approximately 0.8879. This means that around 88.79% of the variance in sales can be explained by the linear relationship.

7. Model Interpretation

- **Feature Importance**:
  - I used lagging sales and previous 3 month average sales that can be used to effectively predict future sales trend.
  - Lagging sales likely capture historical patterns and seasonality, while the 3-month average smooths out short-term fluctuations.

- **Model Insights**:
  - Linear Regression having highest accuracy suggest that company sales are currently growing in a linear way.
  - Steps like better Marketing and Supply Chain Optimization can result in exponential growth for the business, provided its already huge customer base.

8. Model Deployment

- **Deployment Plan**: This model can be used to predict sales for medium scale companies to predict potential sales and relevant expansion strategies with relevant changes and fine tuning.
- **Monitoring and Maintenance**: Services can be offered to companies for free or a minor fee to analyse their data, providing valuable insights while data can be used to in turn enhance the model .

9. Inventory optimization:

I used sevral techniques like EOQ, JIT, ABC Analysis, Safety Stock.

Calculating EOQ (Economic Order Quantity) and JIT (Just-ln-Time)

- **Economic Order Quantity (EQ):** Determines the optimal order quantity to minimize total inventory costs.
- **Just-in-Time (JIT**) Inventory Management Focuses on reducing inventory levels and associated costs by synchronizing inventory with actual demand.
    - Reorder Point (ROP): The inventory level at which a new order should be placed to prevent stockouts. It ensures that stock is replenished before it runs out.
    - Lead Time: The time required from placing an order to receiving it. Effective management of lead time ensures timely replenishment and aligns inventory levels with demand.
    - Daily Demand: The average demand per day, calculated from the total demand rate. It helps in setting appropriate reorder points and determining how much inventory is needed each day.

  Purpose :These components work together in JIT to ensure inventory is available just when needed. minimizing excess stock and reducing carrying costs.

- **ABC Analysis**

ABC Analysis is an inventory management technique used to categorize inventory items into three categories (A, B, and C) based on their importance and value. The goal is to prioritize inventory management efforts on the most critical items.

- Category A: Items with the highest value and typically the lowest quantity. These items are crucial for operations and require strict inventory control and frequent review.
- Category B: Items with moderate value and quantity. These items are important but not as critical as Category A items, requiring regular but less frequent review.
- Category C: Items with the lowest value and highest quantity. These items are less critical and require minimal management effort and infrequent review.

Purpose : Helps allocate resources effectively, improve inventory turnover, and reduce holding costs by focusing management efforts on high-impact items.

- **Calculating Safety Stock**

  Safety Stock is extra inventory kept to prevent stockouts due to variations in demand or delays in supply. It ensures you have enough stock to cover unexpected changes.

  Components:
  - **Z-Score (Z):** This value represents the desired service level. For example, a Z-score of 1.645 is used for a 95% service level, meaning there's a 95% chance that inventory will not be depleted.
  - **Standard Deviation of Demand During Lead Time:** This measures how much demand fluctuates during the lead time. It is calculated by taking the standard deviation of daily demand and adjusting it for the length of the lead time.

  Purpose: To calculate how much extra inventory is needed to avoid running out of stock, even with uncertainties in demand or delivery times.

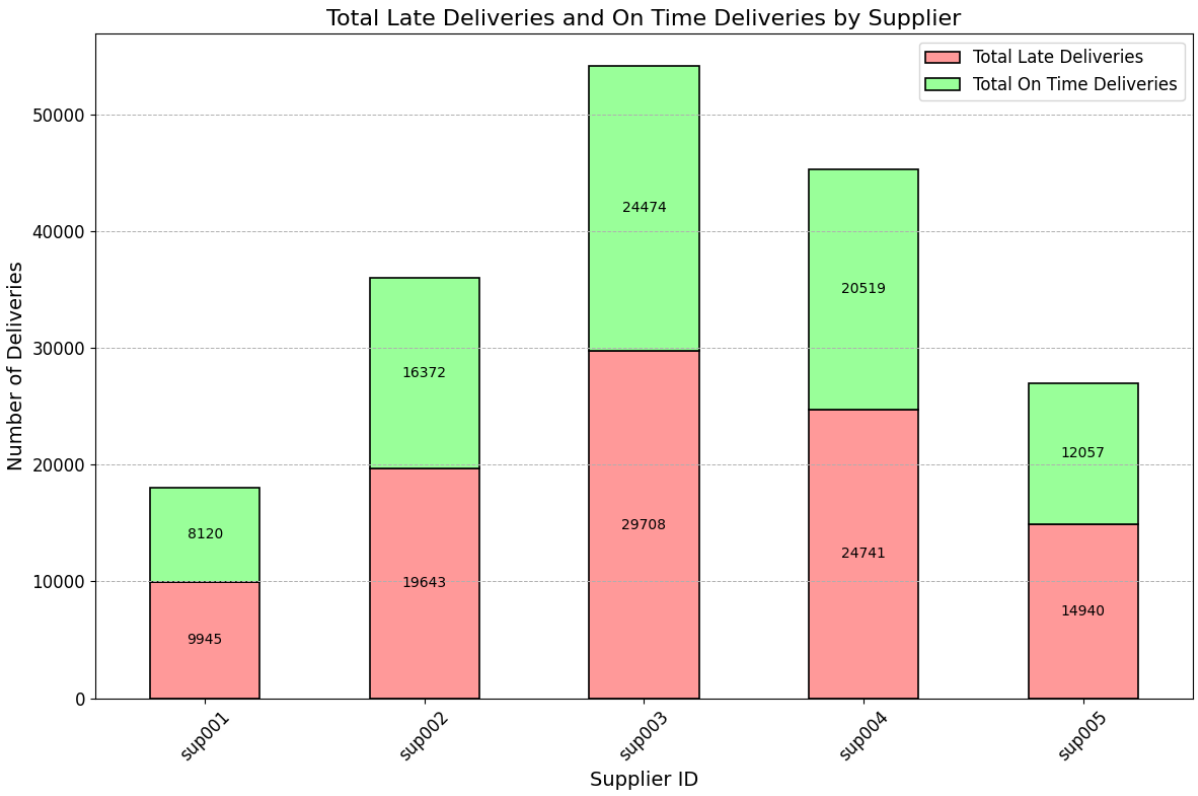
Here are 5 row random sample from dataframe to demonstrate Inventopry optimization

	Product Card Id	Demand_Rate	Product Price	Lead_Time	Sales	Ordering_Cost	Std_Dev_Demand	Holding_Cost	Daily_Demand	Reorder_Point	EOQ	Cumulative_Value	Cumulative_Percentage	Category
81	1059	40	349.99	7	13999.60	209.99	0.00	24.50	0.04	0.25	26.19	36426648.95	99.03	C
51	273	795	27.99	7	22252.05	16.79	1.46	1.96	0.71	4.94	116.74	35871151.53	97.52	C
6	403	22246	129.99	4	2891757.66	77.99	0.00	9.10	19.76	79.03	617.54	28276258.35	76.87	A
45	282	848	31.99	5	27127.52	19.19	1.47	2.24	0.75	3.77	120.57	35731275.21	97.14	C
54	567	859	25.00	5	21475.00	15.00	1.46	1.75	0.76	3.81	121.35	35936558.95	97.69	C

Here are first 5 rows with suggested Safety Stock for each product.

	Product Card Id	Demand_Rate	Product Price	Lead_Time	Sales	Ordering_Cost	Std_Dev_Demand	Holding_Cost	Daily_Demand	Reorder_Point	EOQ	Std_Dev_Demand_Lead_Time	Safety_Stock
0	19	64	124.99	4	7999.36	74.99	0.00	8.75	0.06	0.23	33.12	0.00	0.00
1	24	231	79.99	7	18477.69	47.99	1.31	5.60	0.21	1.44	62.93	3.47	5.72
2	35	65	159.99	6	10399.35	95.99	0.00	11.20	0.06	0.35	33.38	0.00	0.00
3	37	781	34.99	3	27327.19	20.99	1.41	2.45	0.69	2.08	115.71	2.44	4.01
4	44	939	59.99	4	56330.61	35.99	1.46	4.20	0.83	3.34	126.87	2.92	4.80

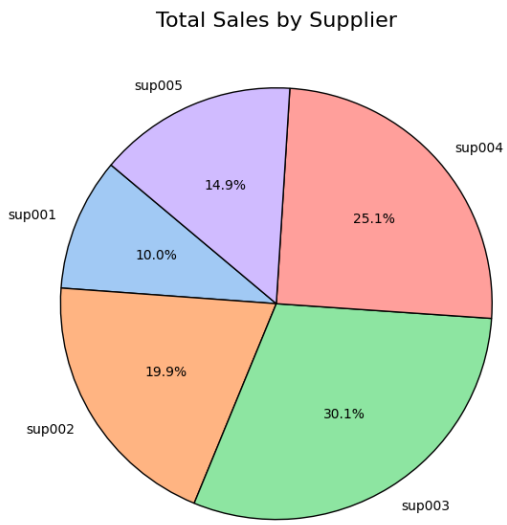
10. Supplier and Transportation Optimization

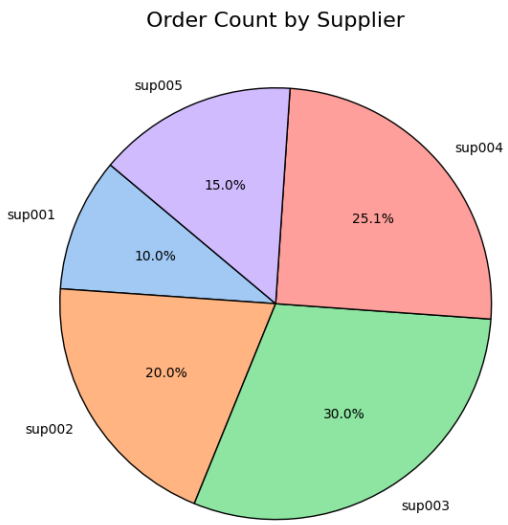


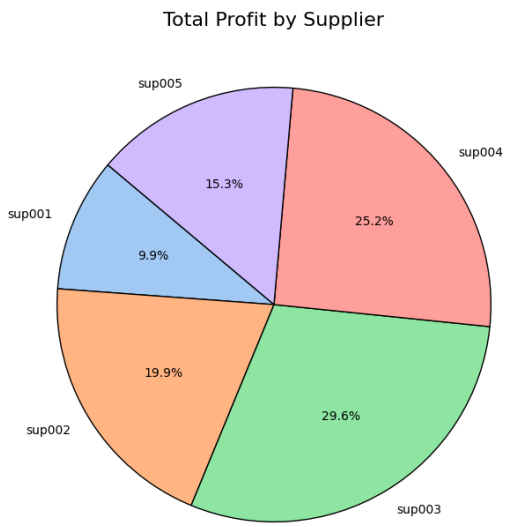
- All suppliers have a high late delivery count, more than half the deliveries are late which creates a bad impression

lier_ID	Total_Transportation_Cost	Avg_Transportation_Cost	Avg_Lead_Time	Total_Sales	Total_Profit	Order_Count	Total_Late_Deliveries	Total_On_Time_Deliveries	Late_delivery_risk	On_Time_Delivery_Rate
up001	246767.623	13.660	6.986	3661680.960	394629.490	18065	9945	8120	0.551	0.449
up002	494724.213	13.737	6.954	7335813.290	790640.372	36015	19643	16372	0.545	0.455
up003	746855.143	13.784	6.983	11081666.291	1174694.150	54182	29708	24474	0.548	0.452
up004	621660.349	13.735	6.958	9228008.807	1000616.781	45260	24741	20519	0.547	0.453
up005	369300.395	13.679	6.973	5477565.666	606322.180	26997	14940	12057	0.553	0.447

- With available data, it is suggested optimize supply chain or look for better suppliers

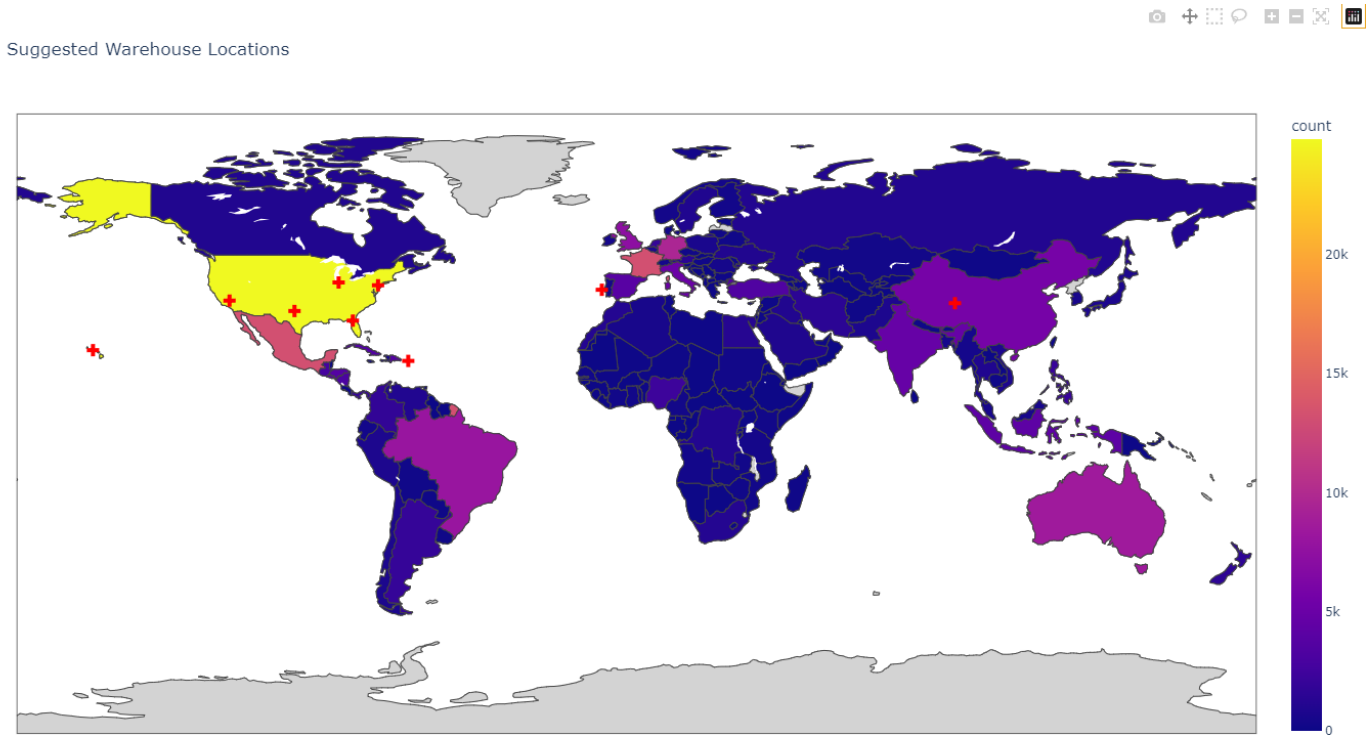






- Profit, Sales and Order count follow similar distribution for all suppliers.

11. Network Design and Optimization



I used K-means clustering to determine Warehouse Locations based on longitudes and latitudes.

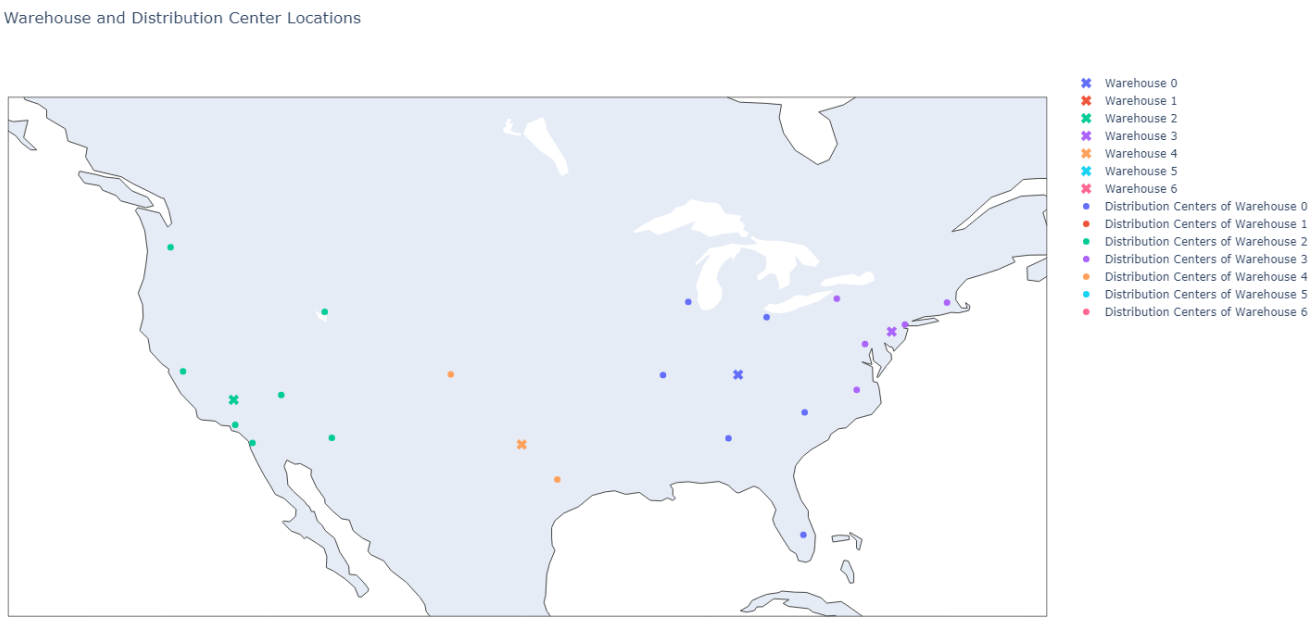
One warehouse is suggested approximately around 20 thousand orders **locations**.

Suggested warehouse and its respective distribution centre location baed on K means clustering

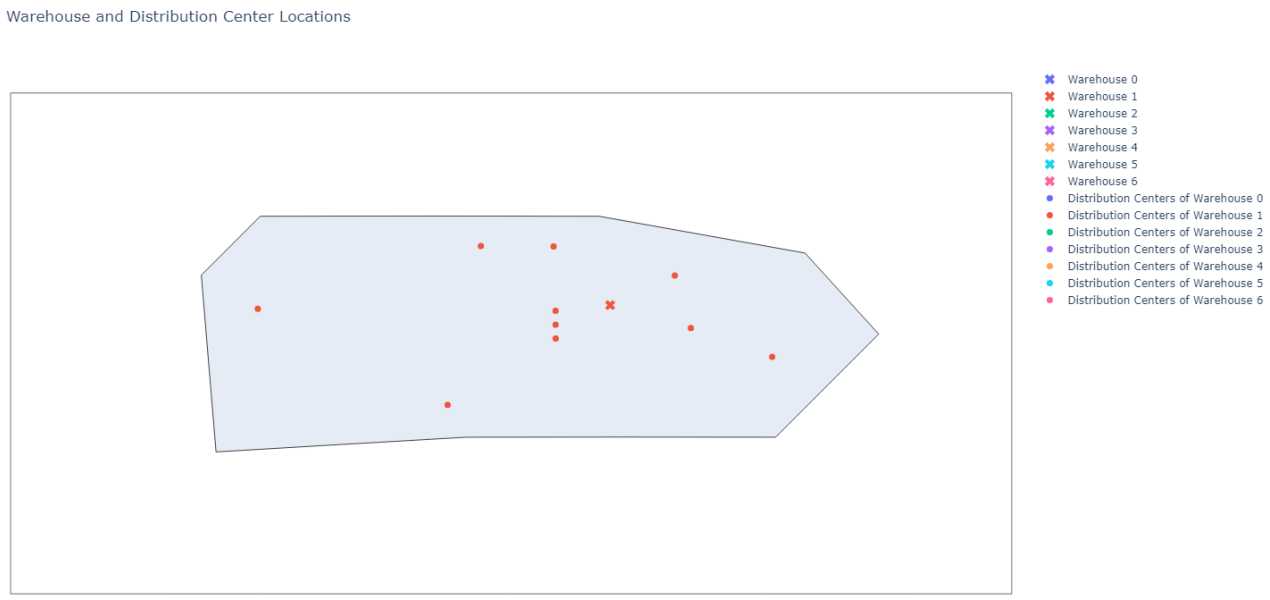
One warehouse is suggested approximately around 20 thousand order locations.

A distribution centre is suggested around 5 thousand order locations

Warehouses and Distribution centres across USA



Warehouses and Distribution centres across Puerto Rico

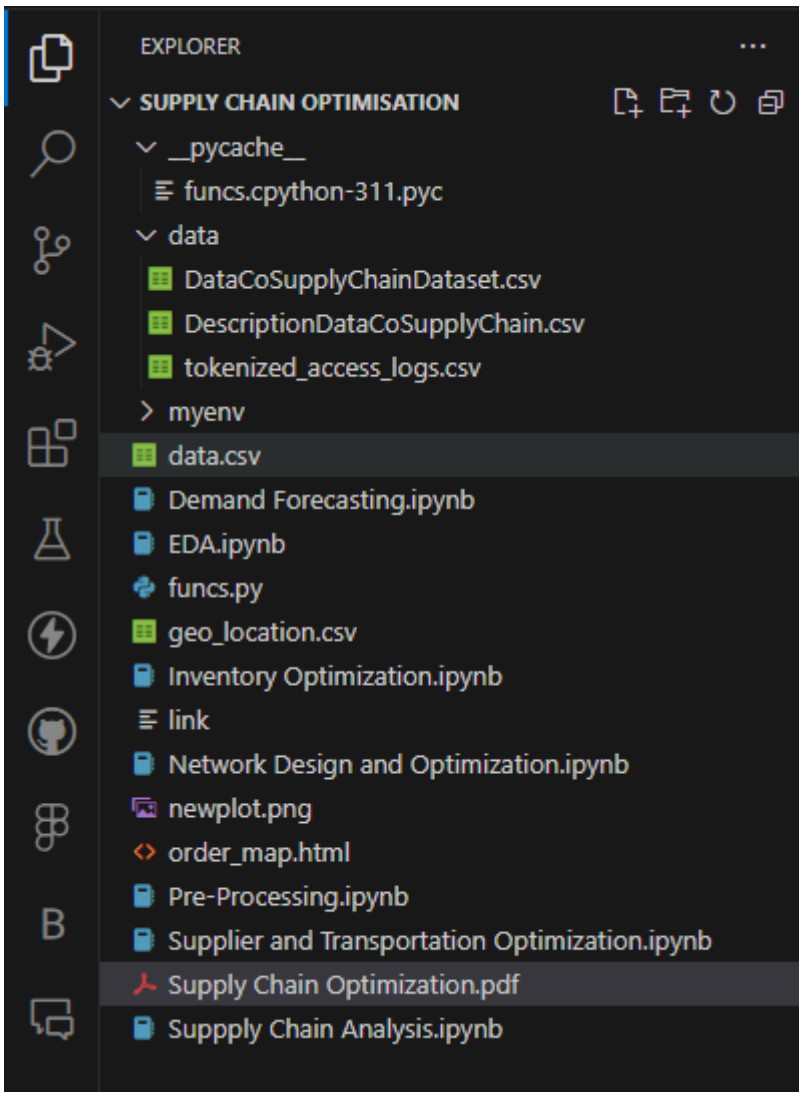


9. Conclusion

- **Summary:**
  - This project successfully optimized the supply chain operations by analyzing current processes, identifying inefficiencies, and developing strategies for improvement.
- **Challenges:**
  - Selecting appropriate data was a significant challenge, requiring thorough evaluation of various datasets to determine the most suitable one.
  - Understanding the project scope was particularly complicated as supply chain optimization was a completely new concept.
  - I had to rush through documentation as I only got time on the last day of submission to make it.
- **Future Work:**
  - Generalized models can be developed to provide supply chain optimization solutions for startups, which often operate on tight budgets and can benefit from a basic overview.
  - Personalized supply chain management can be further refined for larger product-based organizations to enhance efficiency and effectiveness.

10. Appendices

Additional Visualizations:



Project Structure Screenshot

- **References**:
    - ChatGPT
    - BlackBox.ai
    - Windows Co-pilot
    - YouTube
    - GitHub Co-pilot